

Lecture Notes: Graph Theory—Sorted Edges / Cheapest Link

Math F113X Math and Society

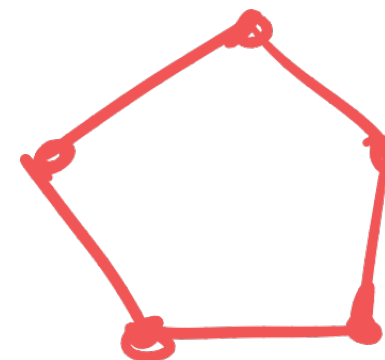
The Sorted Edges / Cheapest Link Algorithm

Steps: Add the next cheapest edge to your circuit **unless**

1. it closes the circuit too soon, or
2. creates a degree 3 vertex.

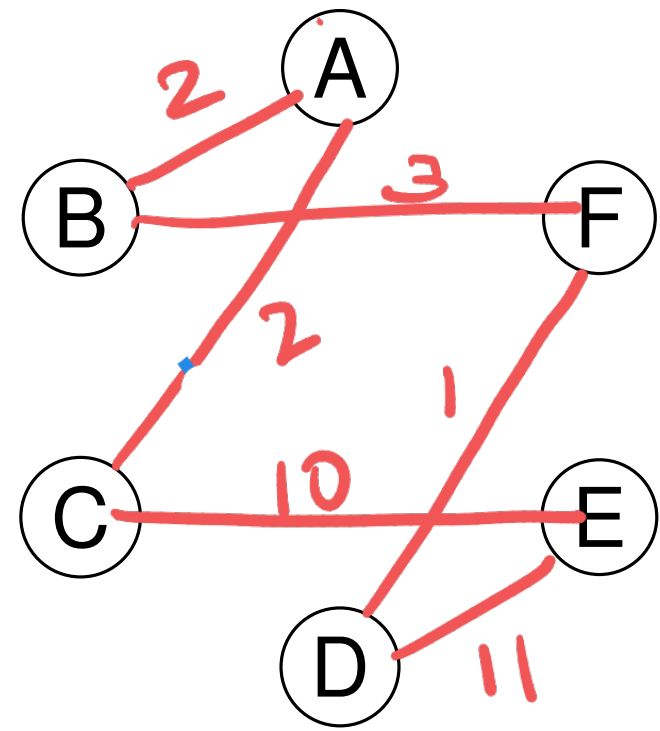
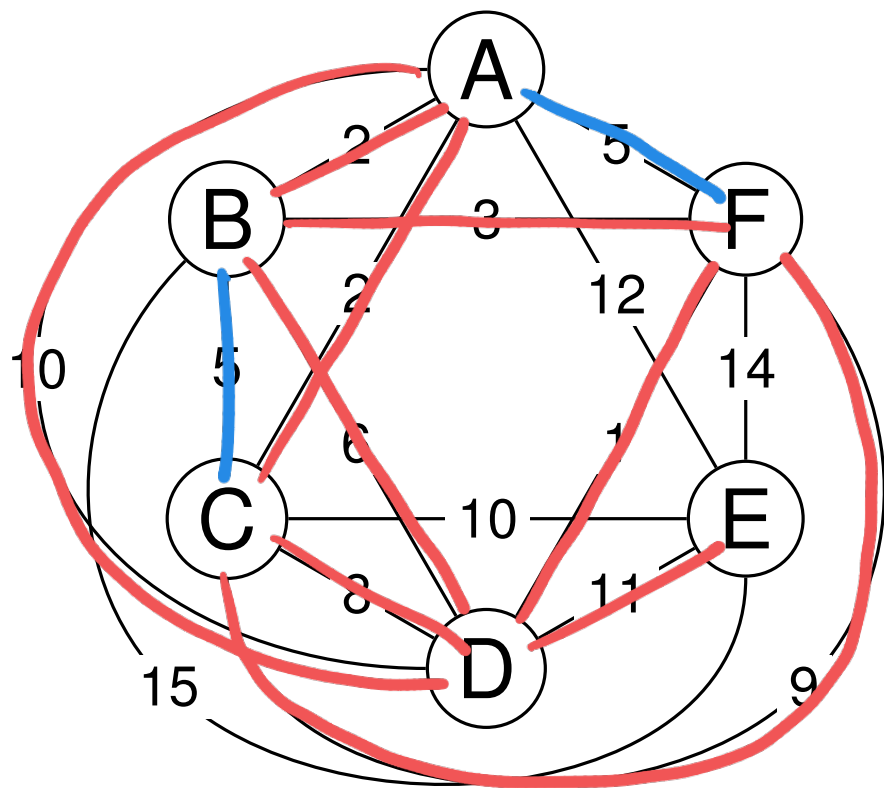
← want to end up with a HC! everything degree 2

Break ties by choosing the alphabetically smallest edge.



Apply the **Sorted Edges** Algorithm

Apply the Sorted Edges Algorithm to find a Hamiltonian circuit. Draw in the edges, labeled with their weight, as you add them on the empty graph.



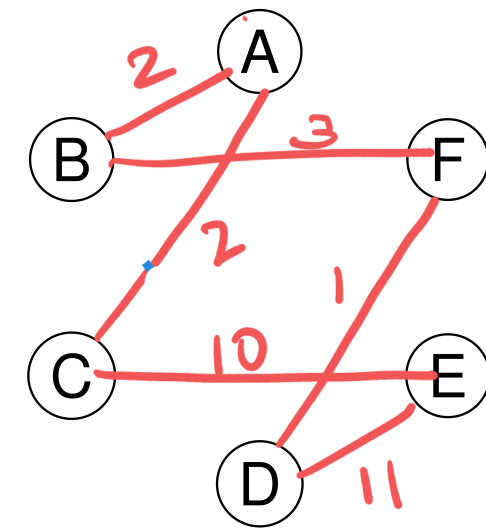
edge	<i>FD</i>	<i>AB</i>	<i>AC</i>	<i>BF</i>	<i>AF</i>	<i>BC</i>	<i>BD</i>	<i>CD</i>	<i>CF</i>	<i>AD</i>	<i>CE</i>	<i>DE</i>	<i>AE</i>	<i>EF</i>	<i>BE</i>
weight	1	2	2	3	5	5	6	8	9	10	10	11	12	14	15
used?	✓	✓	✓	✓	✗	✗	✗	✗	✗	✗	✓	✓	STOP!		

The circuit

List the vertices of the Hamiltonian circuit, starting at vertex A.

A B F D E C A

$$\text{Weight} = 2 + 3 + 1 + 11 + 10 + 2$$



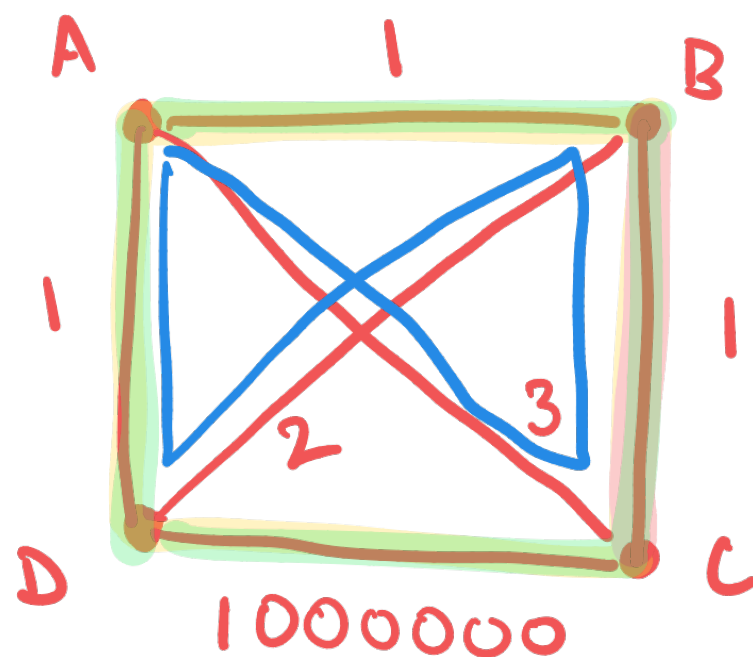
Total weight of the circuit? 29

Do you think the circuit we obtained is the best possible?

maybe?

When does Sorted Edges fail?

Can you construct a graph such that the Sorted Edges Algorithm will never result in a Hamiltonian circuit of smallest weight? What does this tell us about the Sorted Edges Algorithm?



Sorted edges

AB	1
AD	1
BC	1
BD	2
AC	3
CD	1000000

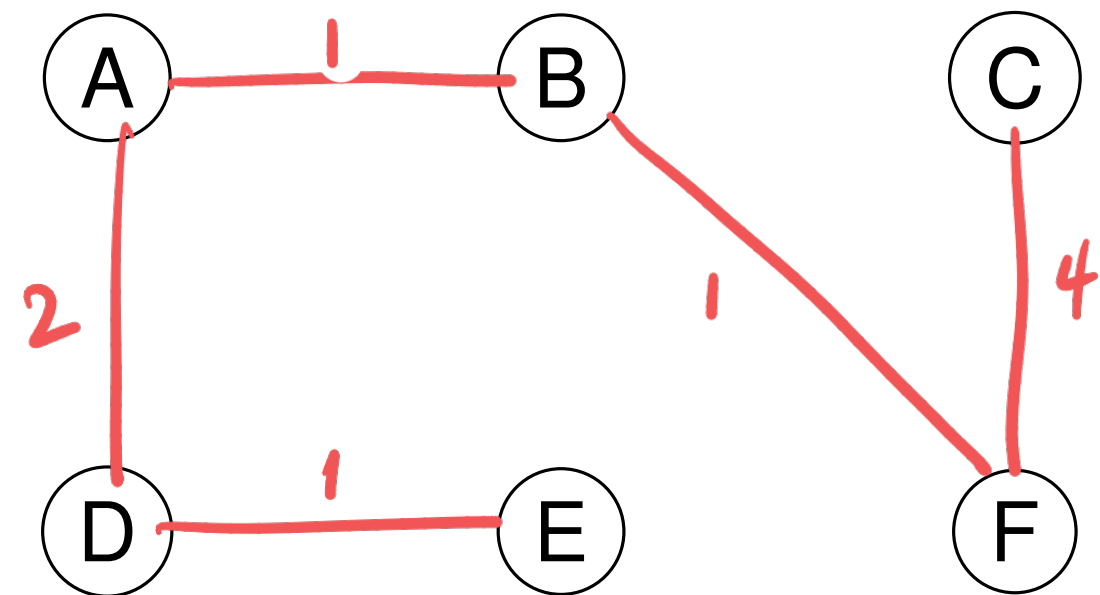
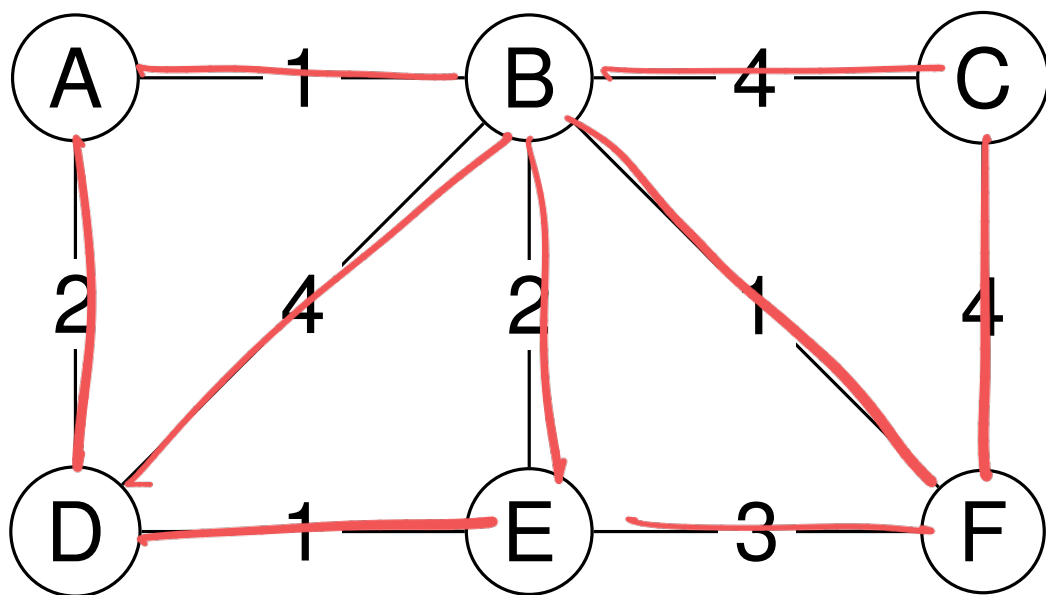


Total weight = 1000003

ACBD A is minimal HC; weight = 7

What happens on this graph?

What happens if you apply Sorted Edges/Cheapest Link to the following graph?

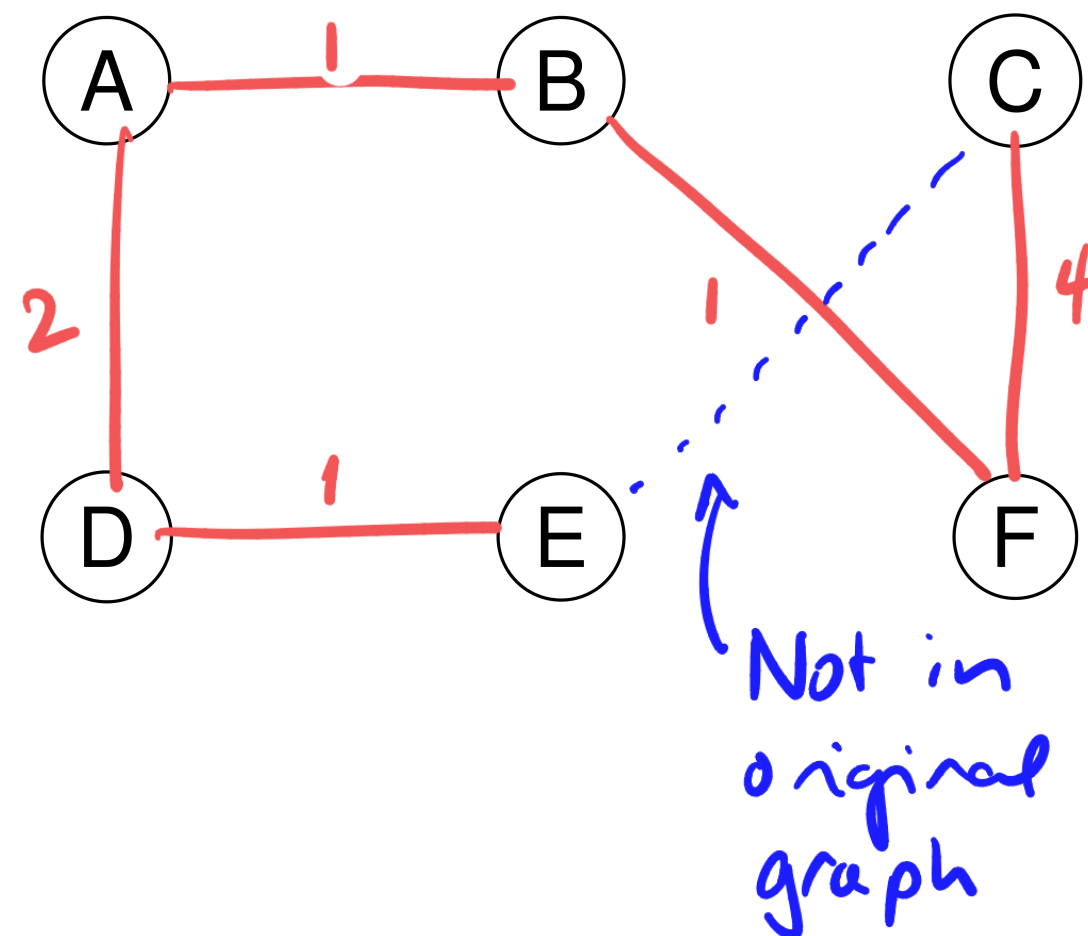


edge	<i>AB</i>	<i>BF</i>	<i>DE</i>	<i>AD</i>	<i>BE</i>	<i>EF</i>	<i>BC</i>	<i>BD</i>	<i>CF</i>
weight	1	1	1	2	2	3	4	4	4
used?	✓	✓	✓	✓	✗	✗	✗	✗	✓

This is not a Hamiltonian circuit!

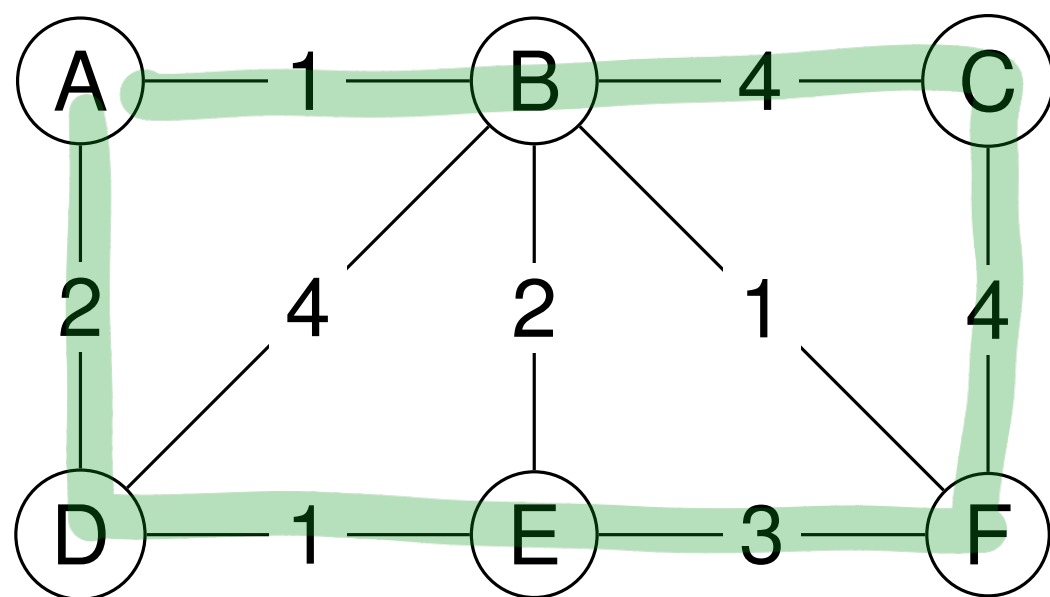
What is the problem here?

We ran out of available edges before we could complete the circuit!



The smallest-weight Hamiltonian circuit

There is a Hamiltonian circuit on this graph. What is the smallest-weight Hamiltonian circuit you can find?



Circuit: ABC FE DA Weight: 15